DECODING DATA SCIENCE

Day 3 | From Chatbot to Agent

SQL on the DDS leave database, three tools, and an agent that decides which one to use.



TIME	YOU WILL BUILD	FILES NEEDED
About 2 hours, hands-on	DecoBot Agent with 3 tools	DDS_Leave_Policy_v2.pdf and dds_leave.db

Three days, one assistant



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One company, one topic: leave at Decoding Data Science.

THE PROGRAMME

DAY 1 | DONE

Control the AI

System prompt and clarity. No files needed.

DAY 2 | DONE

Give it the policy

RAG and chunking over the leave policy PDF.

DAY 3 | TODAY

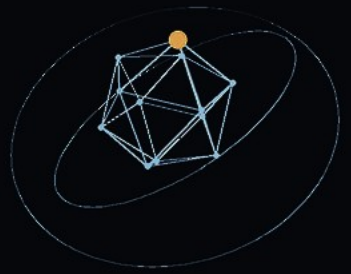
Give it data and tools

SQL, three tools and an agent that decides.

TODAY

DecoBot answers about the rules and about my own numbers, and chooses the right tool by itself.

Today's Plan



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Six steps in about two hours. Each one is a section of the Day 3 notebook.

0:00	**Recap and setup**	Colab secret, upload the PDF and the database
0:15	**Part A: SQL basics**	Five moves on the leave database, you write the queries
0:40	**Part B: three tools**	Turn Python functions into tools with clear descriptions
1:00	**Part C: the agent**	Tools, system prompt and memory in one agent
1:15	**Part D: scenes**	Five scenes: predict which tool it picks
1:40	**Experiment and wrap-up**	Bad tool descriptions, chatbot vs agent, next steps

Rules vs Personal Data



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A PDF knows the rules. It does not know you.

THE RULES

Policy PDF

Notice periods, entitlements, carry-over, certificates. The same for everyone. RAG from Day 2 handles it.

MY NUMBERS

Database

How many days I used, what is booked, what is left, and my past requests. Different for each person. SQL handles it.

THE GOAL

"Can I take 5 days from 14 December?" needs both: the notice rule from the PDF and my balance from the database.

The DDS Leave Database



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One SQLite file, two tables, ready to query in Colab.

TABLE 1

employees

100 rows
employee_id, name, department, job_title,
join_date
annual_total, used, booked, remaining
sick_total, used, remaining

TABLE 2

leave_requests

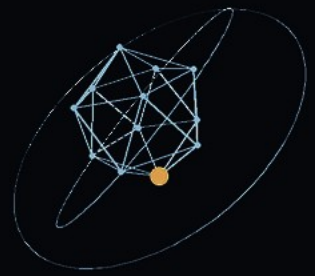
273 rows
request_id, employee_id, leave_type
start_date, end_date, working_days
status, requested_on

THE LINK

employee_id connects the two tables. One employee has many requests.

Five SQL Moves

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You will write these yourself in the notebook. This is all the SQL the agent needs.

SELECT, WHERE	`SELECT name, annual_remaining FROM employees WHERE name = 'Aisha Khan'`
ORDER BY, LIMIT	`SELECT name, sick_used FROM employees ORDER BY sick_used DESC LIMIT 3`
GROUP BY	`SELECT status, COUNT(*) FROM leave_requests GROUP BY status`
JOIN	`SELECT e.name, r.start_date FROM leave_requests r JOIN employees e ON e.employee_id = r.employee_id`
? placeholder	`WHERE name LIKE ?` with the value passed separately

WHY NOT ASK THE AI TO WRITE SQL?

We write the queries inside safe tools. The agent only chooses the tool and fills in a name.

Never Glue Text Into SQL



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User text is data, not part of the query.

```
# WRONG: the text becomes part of the query
con.execute(f"SELECT * FROM employees WHERE name =
'{name}'")

# RIGHT: the ? placeholder keeps it as data
con.execute(
    "SELECT * FROM employees WHERE name LIKE ?",
    (f"%{name}%",))
```

The risk

A name like x' OR '1'='1 would change the meaning of the query.

The habit

Always use placeholders, and make tools read-only.

A Tool Is a Python Function



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The description is what the model reads to decide when to call it.

```
def get_leave_balance(name: str) -> str:
    """Get ONE employee's current leave balance
    (annual and sick) from the HR database.
    Use this when someone asks how many leave days
    they have used, booked or left.
    Input: the employee's full name,
    for example 'Aisha Khan'."""
    ...  # runs the SQL with a ? placeholder

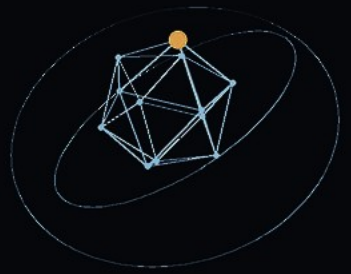
FunctionTool.from_defaults(fn=get_leave_balance)
```

Name
Says what it does.

Docstring
Says when to use it, and what input it needs, with an example.

Type hints
Tell the model what to pass in.

Three Read-Only Tools



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Each tool answers one kind of question.

TOOL 1

get_leave_balance

Annual and sick leave: total, used, booked, remaining.

"How many days do I have left?"

TOOL 2

list_leave_requests

Past and upcoming requests with dates and status.

"What did I book for December?"

TOOL 3

search_leave_policy

The Day 2 policy engine, wrapped as a tool.

"What is the notice for 10 days?"

DESIGN RULE

One tool, one job, one clear description. No tool changes data.

The Agent Loop



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A chatbot answers. An agent decides, acts, looks at the result and decides again.

STEP 1

Think

What does the question need? Rules, data, or both?

STEP 2

Call a tool

Pick a tool and fill in the input, for example the name.

STEP 3

Read the result

The tool output comes back as text for the model.

STEP 4

Answer or repeat

Enough information? Answer. Otherwise call another tool.

IN LLAMAINDEX

`FunctionAgent(tools=tools, llm=llm, system_prompt=SYSTEM_PROMPT)` and `Context(agent)` for memory

The System Prompt Still Matters



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Day 1 comes back: the agent follows the prompt when it picks tools.

IDENTITY

Who is asking?

Ask for the full name first. Discuss only that person's leave.

TOOLS

Which tool when?

Rules go to search_leave_policy. Personal numbers go to the database tools.

BOUNDARIES

What to refuse

No changes to data. No other people's data. Leave topics only.

STYLE

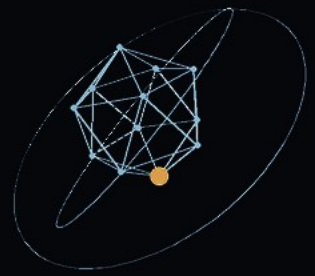
How to answer

Short, clear, and say which source was used.

TODAY'S DATE

The prompt also states the simulation date, 2026-10-28, so the agent can count the days of notice.

Scenes 1 and 2: One Tool Each



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Predict first: which tool will it pick?

SCENE 1 | DATA

Aisha asks about herself

"Hi, I'm Aisha Khan. How many annual leave days do I have left?"

Tool: get_leave_balance
Answer: 2 days (15 used, 5 booked)

SCENE 2 | RULES

A rule question

"What is the notice period if I want 10 or more working days off?"

Tool: search_leave_policy
Answer: 30 calendar days (section 3.4)

WATCH THE LOG

The notebook prints a [tool] line each time the agent calls a tool. That is how you see its decision.

Scene 3: Both Tools, One Answer



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"I want to take 5 working days off from 14 December. Is that possible?"

TOOL: POLICY

Notice: OK

3 to 9 working days need 14 calendar days. 14 December is 47 days away.

TOOL: DATABASE

Balance: not OK

Aisha has 2 annual days left. 5 days were requested.

THE AGENT COMBINES

Not possible

The notice is fine, but the balance is too low. It explains both parts.

WHY THIS MATTERS

No chatbot with only a PDF, or only a database, can answer this. The agent plans two calls and joins the facts.

Scenes 4 and 5: Memory and Boundaries



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A new person, a new conversation, and requests it must refuse.

SCENE 4 | NEW CONTEXT

John Doe, 10 days

ctx = Context(agent) resets the memory.

"10 working days from 7 December."
17 days left, 40 days notice vs 30 needed. OK.

SCENE 5 | REFUSE

Not allowed

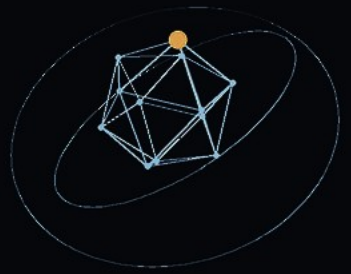
"Cancel my leave request."
No write tool exists.

"Show me Maria Garcia's balance."
Only your own leave.

TWO DEFENCES

The prompt says no. And the tools cannot do it anyway. Safe design uses both.

Bad Tool Descriptions



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Same agent, same questions, but the descriptions say only "Useful tool."

BEFORE

Clear descriptions

"Use this when someone asks how many leave days they have used, booked or left. Input: full name."

AFTER

"Useful tool."

"Another tool." "A tool."

The model has nothing to decide with.

WHAT YOU WILL SEE

Wrong or no tool

Wrong tool, missing name, or a guess instead of a lookup. Compare with the good agent.

LESSON

The agent is only as good as its tool descriptions. Write them like instructions for a new colleague.

Chatbot vs Agent

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Same model, three days of added capability.



Day 1	**Chatbot with a system prompt**	Follows rules, asks when unclear. Knows only the prompt.
Day 2	**RAG assistant**	Answers from the policy PDF with sections. Knows the rules.
Day 3	**Agent with tools**	Reads the question, picks tools, combines facts, answers.

THE PATTERN

Prompt for behaviour. RAG for knowledge. Tools for live data and actions.

Where to Go Next



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Not for this bootcamp, but the natural path to a real product.

SECURITY

Real login

Identify the user with company sign-in, not with a typed name.

ACTIONS

Write tools

Submit a request, but only after the user confirms.

QUALITY

Evaluation set

Fixed test questions, scored after every change.

SCALE

More documents

Add other HR policies. Keep chunking tested.

OPERATIONS

Logging

Record every tool call, then put it behind a web app.

THREE BUILDING BLOCKS

A clear system prompt, a retrieval engine that you measured, and tools with good descriptions. You built all three.